The documentation and process conversion measures necessary to comply with this revision shall be completed by 22 September 2001.

INCH-POUND

MIL-PRF-19500/323G
22 June 2001
SUPERSEDING
MIL-PRF-19500/323F
15 April 1998

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, SWITCHING
TYPES 2N3250A, 2N3251A, 2N3250AUB, 2N3251AUB,
JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for PNP silicon switching transistors. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500. Two levels of product assurance are provided for die.

1.2 Physical dimensions. See figures 1 (similar to TO-18), 2 (surface mount), and 3 (die) herein.

1.3 Maximum ratings.

| $P_T$ (1) $T_A$ = +25°C | $V_{CBO}$ | $V_{CEO}$ | $V_{EBO}$ | $I_C$ | $T_{OP}$ and $T_{STG}$ | $R_{\theta JA}$ (1) |
|---|---|---|---|---|---|---|
| W | V dc | V dc | V dc | mA dc | °C | °C/W |
| 0.36 | 60 | 60 | 5.0 | 200 | -65 to +200 | 417 |

(1) Derate linearly 2.4 mW/°C above $T_A$ = +25°C.

1.4 Primary electrical characteristics.

| Limits | $h_{FE1}$ $V_{CE}$ = 1.0 V dc $I_C$ = 0.1 mA dc | | $h_{FE3}$ (1) $V_{CE}$ = 1.0 V dc $I_C$ = 10 mA dc | | $h_{FE4}$ (1) $V_{CE}$ = 1.0 V dc $I_C$ = 50 mA dc | | $\|h_{fe}\|$ f = 100 MHz $V_{CE}$ = 20 V dc; $I_C$ = 10 mA dc | |
|---|---|---|---|---|---|---|---|---|
| | Min | Max | Min | Max | Min | Max | Min | Max |
| 2N3250A | 40 | | 50 | 150 | 15 | | 2.5 | 9.0 |
| 2N3251A | 80 | | 100 | 300 | 30 | | 3.0 | 9.0 |

(1) Pulsed (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center, Columbus, ATTN: DSCC/VAC, Post Office Box 3990, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

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1.4 Primary electrical characteristics - Continued.

| Limits | $r_b'C_C$<br>$V_{CE}$ = 20 V dc<br>$I_C$ = 10 mA dc<br>f = 31.8 MHz | $V_{CE(SAT)1}$<br>$I_C$ = 10 mA dc<br>$I_B$ = 1.0 mA dc | $C_{obo}$<br>$V_{CB}$ = 10 V dc<br>$I_E$ = 0<br>100 kHz ≤ f ≤ 1 MHz | $t_{on}$<br>$I_C$ = 10 mA dc<br>$I_B$ = 1.0 mA dc | $t_{off}$<br>$I_C$ = 10 mA dc<br>$I_B$ = 1.0 mA dc | | $N_F$<br>$V_{CE}$ = 5 V dc<br>$I_C$ = .1 mA dc<br>Rg = 1kΩ<br>f = 100 Hz |
|---|---|---|---|---|---|---|---|
| | | | | | 2N3250A | 2N3251A | |
| | ps | V dc | pF | ns | ns | ns | dB |
| Min | 5 | | | | | | |
| Max | 250 | 0.25 | 6 | 70 | 250 | 300 | 6 |

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

DEPARTMENT OF DEFENSE

MIL-STD-750 - Test Methods for Semiconductor Devices.

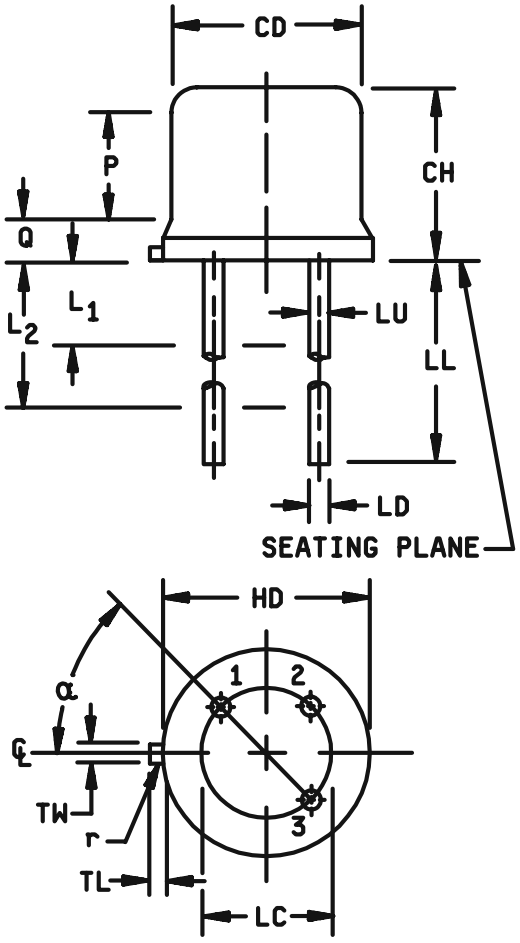
(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Document Automation and Production Services (DAPS), Building 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.



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| Symbol | Dimensions | | | | Notes |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| CD | .178 | .195 | 4.52 | 4.95 | |
| CH | .170 | .210 | 4.32 | 5.34 | |
| HD | .209 | .230 | 5.31 | 5.84 | |
| LC | .100 TP | | 2.54 TP | | 6 |
| LD | .016 | .021 | 0.41 | 0.53 | 7, 8 |
| LL | .500 | .750 | 12.70 | 19.05 | 7,. 8, 12 |
| LU | .016 | .019 | 0.41 | 0.48 | 7, 8 |
| $L_1$ | | .050 | | 1.27 | 7, 8 |
| $L_2$ | .250 | | 6.35 | | 7, 8 |
| Q | | .040 | | 1.02 | 5 |
| r | | .007 | | 0.178 | 10 |
| TL | .028 | .048 | 0.71 | 1.22 | 3, 4 |
| TW | .036 | .046 | 0.91 | 1.17 | 3 |
| α | 45° TP | | 45° TP | | 6 |

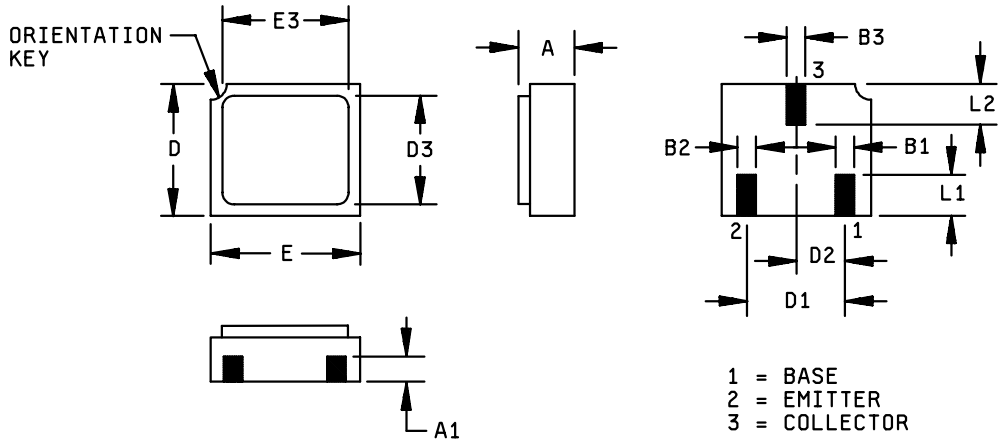



NOTES:
1. Dimension are in inches.
2. Metric equivalents are given for general information only.
3. Beyond r (radius) maximum, TW shall be held for a minimum length of .011 (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods.
7. Dimension LU applies between $L_1$ and $L_2$. Dimension LD applies between $L_2$ and LL minimum. Diameter is uncontrolled in $L_1$ and beyond LL minimum.
8. All three leads.
9. The collector shall be internally connected to the case.
10. Dimension r (radius) applies to both inside corners of tab.
11. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions (similar to TO-18 ).



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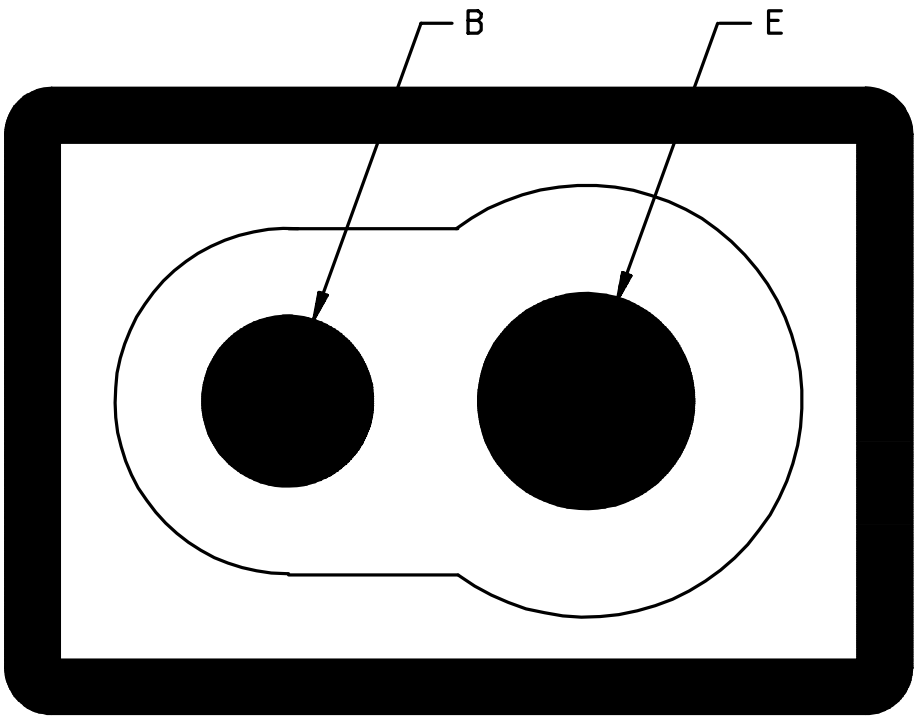

| Symbol | Dimensions | | | | Note |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| A | .046 | .056 | 0.97 | 1.42 | |
| A1 | .017 | .035 | 0.43 | 0.89 | |
| B1 | .016 | .024 | 0.41 | 0.61 | |
| B2 | .016 | .024 | 0.41 | 0.61 | |
| B3 | .016 | .024 | 0.41 | 0.61 | |
| D | .085 | .108 | 2.41 | 2.74 | |
| D1 | .071 | .079 | 1.81 | 2.01 | |
| D2 | .035 | .039 | 0.89 | 0.99 | |
| D3 | .085 | .108 | 2.41 | 2.74 | |
| E | .115 | .128 | 2.82 | 3.25 | |
| E3 | | .128 | | 3.25 | |
| L1 | .022 | .038 | 0.56 | 0.96 | |
| L2 | .022 | .038 | 0.56 | 0.96 | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Physical dimensions, surface mount (UB version).



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NOTES:

1. Chip size ..... 15 x 19 mils ± 1 mil
2. Chip thickness ..... 10 ± 1.5 mil
3. Top metal ..... Aluminum 15,000Å minimum, 18,000Å nominal
4. Back metal ..... A. Gold 2,500Å minimum, 3,000Å nominal
   B. Eutectic Mount – No Gold
5. Backside ..... Collector
6. Bonding pad ..... B = 3 mils, E = 4 mils diameter
7. Passivation ..... $Si_3N_4$ (Silicon Nitride) 2 kÅ min, 2.2 kÅ nom.

FIGURE 3. Physical dimensions, JANHCA and JANKCA die.



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3. REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-19500.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

$I_{BEX}$ - - - Base cutoff current (dc) with specified circuit between the collector and emitter.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1 (TO-18), figure 2 (UB surface mount), and figure 3 (die) herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characterics are as specified in 1.3, 1.4, and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3 herein.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).



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4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.3 Screening (JANS, JANTX and JANTXV levels only). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see table IV of MIL-PRF-19500) | Measurement | |
|---|---|---|
| | JANS level | JANTX and JANTXV levels |
| 9 | $h_{FE3}$, $I_{CBO2}$ | Not applicable |
| 11 | $I_{CBO2}$; $h_{FE3}$;<br>$\Delta I_{CBO2}$ = 100 percent of initial value or 5 nA dc, whichever is greater,<br>$\Delta h_{FE3}$ = 25 percent change from initial value | $I_{CBO2}$ and $h_{FE3}$ |
| 12 | See 4.3.1 | See 4.3.1 |
| 13 | Subgroups 2 and 3 of table I herein;<br>$\Delta I_{CBO2}$ = 100 percent of initial value or 5 nA dc, whichever is greater;<br>$\Delta h_{FE3}$ = 25 percent change from initial value. | Subgroup 2 of table I herein;<br>$\Delta I_{CBO2}$ = 100 percent of initial value or 5 nA dc, whichever is greater;<br>$\Delta h_{FE3}$ = 25 percent change from initial value. |

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: $T_A$ = Room ambient as defined in 4.5 of MIL-STD-750; $V_{CB}$ = 10- 30 V dc (10 V dc for JANS); $P_T$ = 360 mW. NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.3.2 Screening JANC. Screening shall be in accordance with MIL-PRF-19500. As a minimum, die shall be 100 percent probed to ensure compliance with group A, subgroup 2.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table V of MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) and table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with group A, subgroup 2 herein. Delta requirements shall be in accordance with the steps of table III herein as specified in the notes for table III.



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4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

| Subgroup | Method | Conditions |
|---|---|---|
| B4 | 1037 | $V_{CB}$ = 10 V dc. |
| B5 | 1027 | $V_{CB}$ = 10 - 30 V dc; $T_A$ = +125°C ±25°C for 96 hours with $P_T$ adjusted according to the chosen $T_A$ to give $T_J$ = +275°C minimum. Optionally, the test may be conducted for minimum 216 hrs with $P_T$ adjusted to achieve $T_J$ = +225°C minimum, sample size (for option) = 45, c = 0. In this case, the ambient temperature shall be adjusted such that a minimum 75 percent of maximum rated $P_T$ (see 1.3) is applied to the device under test. (NOTE: If a failure occurs, resubmission shall be at the test conditions of the original sample.) |
| B6 | 3131 | See 4.5.3 herein. |

4.4.2.2 Group B inspection, table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

| Subgroup | Method | Conditions |
|---|---|---|
| B3 | 1027 | $V_{CB}$ = 10 - 30 V dc; $T_A$ = room ambient as defined in the general requirements of MIL-STD-750. Power shall be applied to achieve $T_J$ = +150°C minimum and a minimum power dissipation $P_D \geq 75$ percent of maximum rated $P_T$ as defined in 1.3. |
| B5 | 3131 | See 4.5.3. |

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with group A, subgroup 2 herein. Delta requirements shall be in accordance with the steps of table III herein as specified in the notes for table III.

4.4.3.1 Group C inspection, table VII of MIL-PRF-19500.

| Subgroup | Method | Conditions |
|---|---|---|
| C2 | 2036 | Test condition E, not applicable to surface mount. |
| C6 | 1027 | $V_{CB}$ = 10 - 30 V dc; $T_A$ = room ambient as defined in the general requirements of MIL-STD-750. Power shall be applied to achieve $T_J$ = +150°C minimum and a minimum power dissipation $P_D \geq 75$ percent of maximum rated $P_T$ as defined in 1.3. |



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4.4.4 Group E Inspection. Group E inspection shall be performed for qualification or re-qualification only. The tests specified in table II herein must be performed to maintain qualification.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Collector - base time constant. This parameter may be determined by applying an rf signal voltage of 1.0 volt (rms) across the collector-base terminals, and measuring the ac voltage drop ($V_{eb}$) with a high impedance rf voltmeter across the emitter-base terminals. With f = 31.8 MHz used for the 1.0 V signal, the following computation applies; $r_b'C_c$ (ps) = 5 x $V_{eb}$ (millivolts), see figure 4.

4.5.3 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with test method 3131 of MIL-STD-750. The following details shall apply:

a. Minimum collector magnitude shall be 36 mA dc.

b. Collector to emitter voltage magnitude shall be 10 V dc.

c. Reference point temperature shall be $+25°C \leq T_R \leq +35°C$. The chosen reference temperature shall be recorded before the test is started.

d. Maximum $R_{\theta JA}$ limit shall be 417°C/W.



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TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical examination | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Collector to base cutoff current | 3036 | Bias condition D; $V_{CB}$ = 60 V dc | $I_{CBO1}$ | | 10 | µA dc |
| Emitter to base cutoff current | 3026 | Bias condition D; $V_{EB}$ = 5 V dc | $I_{EBO}$ | | 10 | µA dc |
| Breakdown voltage collector - emitter | 3011 | Bias condition D; $I_C$ = 10 mA dc; pulsed (see 4.5.1) | $V_{(BR)CEO}$ | 60 | | V dc |
| Collector - base cutoff current | 3036 | Bias condition D; $V_{CB}$ = 40 V dc | $I_{CBO2}$ | | 20 | nA dc |
| Collector - emitter cutoff current | 3041 | Bias condition A; $V_{BE}$ = 3.0 V dc, $V_{CE}$ = 40 V dc | $I_{CEX1}$ | | 20 | nA dc |
| Base cutoff current | 3041 | Bias condition A; $V_{BE}$ = 3.0 V dc; $V_{CE}$ = 40 V dc | $I_{BEX}$ | | 50 | nA dc |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 1.0 V dc; $I_C$ = 0.1 mA dc | $h_{FE1}$ | | | |
| 2N3250A, 2N3250AUB | | | | 40 | | |
| 2N3251A, 2N3251AUB | | | | 80 | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 1.0 V dc; $I_C$ = 1.0 mA dc | $h_{FE2}$ | | | |
| 2N3250A, 2N3250AUB | | | | 45 | | |
| 2N3251A, 2N3251AUB | | | | 90 | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 1.0 V dc; $I_C$ = 10 mA dc, pulsed (see 4.5.1) | $h_{FE3}$ | | | |
| 2N3250A, 2N3250AUB | | | | 50 | 150 | |
| 2N3251A, 2N3251AUB | | | | 100 | 300 | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 1.0 V dc; $I_C$ = 10 mA dc, pulsed (see 4.5.1) | $h_{FE4}$ | | | |
| 2N3250A, 2N3250AUB | | | | 15 | | |
| 2N3251A, 2N3251AUB | | | | 30 | | |

See footnote at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 2 - Continued | | | | | | |
| Current gain linearity | | $\frac{\lvert h_{FE3} - h_{FE1} \rvert}{h_{FE3}} \times 100$ | $h_{FE}$ | | | |
| 2N3250A, 2N3250AUB | | | | | 40 | % |
| 2N3251A, 2N3251AUB | | | | | 30 | % |
| Collector - emitter saturated voltage | 3071 | $I_C$ = 10 mA dc; $I_B$ = 1.0 mA dc | $V_{CE(SAT)1}$ | | 0.25 | V dc |
| Collector - emitter saturated voltage | 3071 | $I_C$ = 50 mA dc; $I_B$ = 5.0 mA dc; pulsed (see 4.5.1) | $V_{CE(SAT)2}$ | | 0.50 | V dc |
| Base - emitter saturated voltage | 3066 | Test condition A; $I_C$ = 10 mA dc; $I_B$ = 1.0 mA dc | $V_{BE(SAT)1}$ | 0.60 | 0.90 | V dc |
| Base - emitter saturated voltage | 3066 | Test condition A; $I_C$ = 50 mA dc; $I_B$ = 5.0 mA dc; pulsed (see 4.5.1) | $V_{BE(SAT)2}$ | | 1.20 | V dc |
| Subgroup 3 | | | | | | |
| High-temperature operation: | | $T_A$ = +150°C | | | | |
| Collector - emitter cutoff current | 3041 | Bias condition A; $V_{CE}$ = 40 V dc; $V_{BE}$ = 3.0 V dc | $I_{CEX2}$ | | 20 | µA dc |
| Low-temperature operation: | | $T_A$ = -55°C | | | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 1.0 V dc; $I_C$ = 1.0 mA dc | $h_{FE5}$ | | | |
| 2N3250A, 2N3250AUB | | | | 20 | | |
| 2N3251A, 2N3251AUB | | | | 40 | | |
| Subgroup 4 | | | | | | |
| Small-signal short-circuit forward-current transfer ratio | 3206 | $V_{CE}$ = 10 V dc; $I_C$ = 1 mA dc; f = 1 kHz | $h_{fe}$ | | | |
| 2N3250A, 2N3250AUB | | | | 50 | 200 | |
| 2N3251A, 2N3251AUB | | | | 100 | 400 | |

See footnote at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 4 - Continued | | | | | | |
| Magnitude of common emitter small-signal short-circuit forward-current transfer ratio | 3306 | $V_{CE}$ = 20 V dc; $I_C$ = 10 mA dc; f = 100 MHz | $\lvert h_{fe} \rvert$ | | | |
| 2N3250A, 2N3250AUB | | | | 2.5 | 9.0 | |
| 2N3251A, 2N3251AUB | | | | 3.0 | 9.0 | |
| Open circuit output capacitance | 3236 | $V_{CB}$ = 10 V dc; $I_E$ = 0<br>100 kHz ≤ f ≤ 1 MHz | $C_{obo}$ | | 6 | pF |
| Input capacitance (output open-circuited) | 3240 | $V_{EB}$ = 1.0 V dc; $I_C$ = 0;<br>100 kHz ≤ f ≤ 1 MHz | $C_{ibo}$ | | 8 | pF |
| Collector - base time constant | | $V_{CE}$ = 20 V dc; $I_C$ = 10 mA dc; f = 31.8 MHz; (see 4.5.2 and figure 4) | $r_b'C_c$ | 5 | 250 | ps |
| Noise figure | 3246 | $V_{CE}$ = 5.0 V dc; $I_C$ = 100 µA dc; Rg = 1 kΩ; f = 100 Hz | NF | | 6 | dB |
| Pulse response: | | | | | | |
| On-time | 3251 | Test condition A; $I_C$ = 10 mA dc; $I_{B1}$ = 1.0 mA dc; (see figure 5) | $t_{on}$ | | 70 | ns |
| Off time | 3251 | Test condition A; $I_C$ = 10 mA dc; $I_{B1}$ = $I_{B2}$ = 1.0 mA dc (see figure 6) | $t_{off}$ | | | |
| 2N3250A, 2N3250AUB | | | | | 250 | ns |
| 2N3251A, 2N3251AUB | | | | | 300 | ns |
| Small-signal open circuit reverse-voltage transfer ratio | 3211 | $V_{CE}$ = 10 V dc; $I_C$ = 1.0 mA dc; f = 1 kHz | $h_{re}$ | | | |
| 2N3250A, 2N3250AUB | | | | | 10 | x $10^{-4}$ |
| 2N3251A, 2N3251AUB | | | | | 20 | x $10^{-4}$ |
| Small-signal short circuit input impedance | 3201 | $V_{CE}$ = 10 V dc; $I_C$ = 1.0 mA dc; f = 1 kHz | $h_{ie}$ | | | |
| 2N3250A, 2N3250AUB | | | | 1 | 6 | kΩ |
| 2N3251A, 2N3251AUB | | | | 2 | 12 | kΩ |
| Small-signal open circuit output admittance | | $V_{CE}$ = 10 V dc; $I_C$ = 1.0 mA dc; f = 1 kHz | $h_{oe}$ | | | |
| 2N3250A, 2N3250AUB | | | | 4 | 40 | µmhos |
| 2N3251A, 2N3251AUB | | | | 10 | 60 | µmhos |

1/ For sampling plan, see MIL-PRF-19500.



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TABLE II. Group E inspection (all quality levels) - For qualification only

| Inspection | MIL-STD-750 | | Qualification |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 12 devices<br>c = 0 |
| Temperature cycling (air to air) | 1051 | Test condition C, 500 cycles | |
| Hermetic seal | 1071 | | |
| Fine leak<br>Gross leak | | | |
| Electrical measurements | | See group A, subgroup 2 herein. | |
| Subgroup 2 | | | 45 devices<br>c = 0 |
| Intermittent life | 1037 | Intermittent operation life: $V_{CB}$ = 10 V dc , 6,000 cycles, $\Delta T_J \geq +100°C$; forced air cooling allowed on cooling cycle only. | |
| Electrical measurements | | See group A, subgroup 2 herein. | |
| Subgroup 3 | | | |
| Not applicable | | | |
| Subgroup 4 | | | |
| Not applicable | | | |
| Subgroup 5 | | | |
| Not applicable | | | |



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TABLE III. Group B and Group C delta measurements. 1/ 2/ 3/

| Step | Inspection | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | | Method | Conditions | | Min | Max | |
| 1. | Forward-current transfer ratio | 3076 | $V_{CE}$ = 1.0 V dc; $I_C$ = 10 mA dc; pulsed (see 4.5.1) | $\Delta h_{FE3}$ | ± 25 percent change from initial value. | | |
| 2. | Collector - base cutoff current | 3036 | Bias condition D; $V_{CB}$ = 40 V dc | $\Delta I_{CBO2}$ | 100 percent of initial value or 5 nA dc, whichever is greater. | | |
| 3. | Collector - emitter voltage (saturated) | 3071 | $I_C$ = 50 mA dc; $I_B$ = 5.0 mA dc | $\Delta V_{CE(Sat)2}$ | 50 mV dc change from initial value. | | |

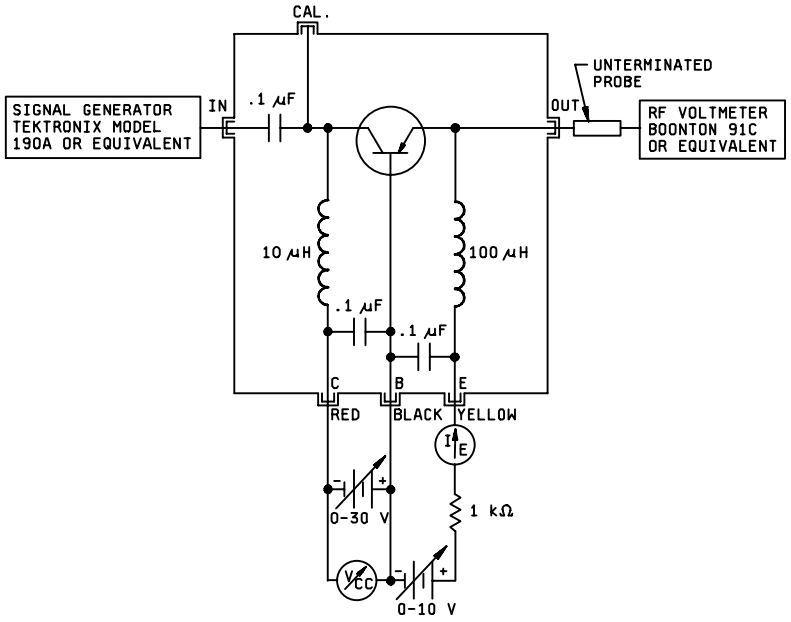
1/ The delta measurements for table VIa (JANS) of MIL-PRF-19500 are as follows:
   a. Subgroup 4, see table III herein, step 3.
   b. Subgroup 5, see table III herein, steps 1, 2, and 3.

2/ The delta measurements for table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500 are as follows: Subgroups 3 and 6, see table III herein, step1.

3/ The delta measurements for table VII of MIL-PRF-19500 are as follows: Subgroup 6, see table III herein, steps 1 and 2 (for JANS) and 1 (for JAN, JANTX, and JANTXV).



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Procedure:

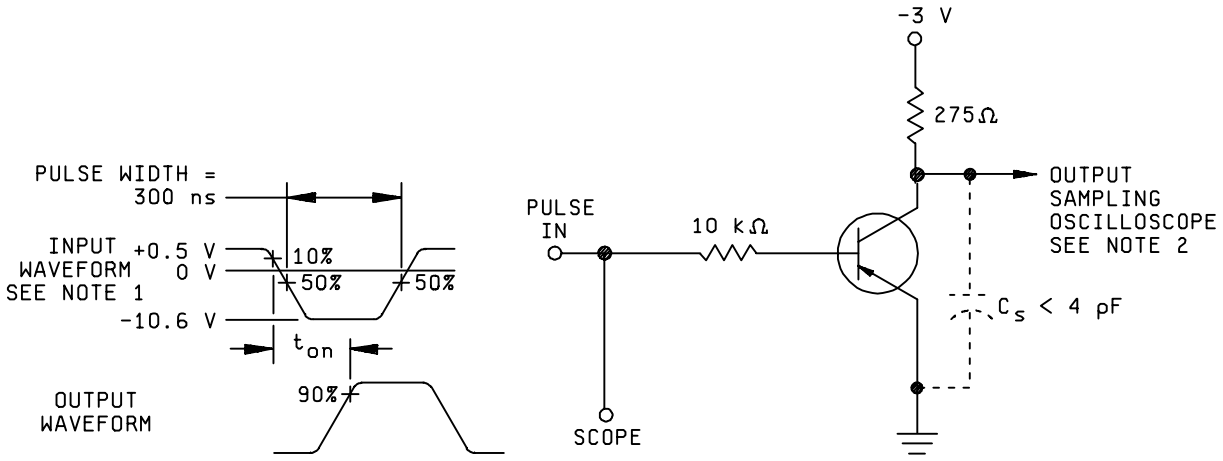
1. Set signal generator to 31.8 MHz and connect to "IN" connector on test jig.
2. Connect low voltage dc power supplies as shown. A 1 K ohm resistor should be placed in series with the emitter power supply to prevent damage to transistors being tested.
3. Set collector supply for $V_{CE}$ = -20 V dc, and emitter supply for $I_C$ = -10 mA.
4. Connect RF voltmeter with unterminated probe adapter to "CAL" connector on test jig. Adjust signal generator until RF voltage is 1 volt (NOTE: Decade switching of voltmeter should be accurate from 1 mV to 3 volts. If not, input voltage may be set using voltage dividers, utilizing lower scales of the RF voltmeter. If this is done, the voltage dividers should be left in place when the voltmeter is removed, as they constitute a load on the input of the circuit.
5. Remove RF voltmeter from "CAL" connector and connect to "OUT" connector. Meter will now read $r_b'C_c$ as follows:

| Meter range full scale |
|---|
| 3 mV |
| 10 mV |
| 30 mV |
| .1 volt |

FIGURE 4. Collector-base time constant test circuit (an equivalent circuit may be used).



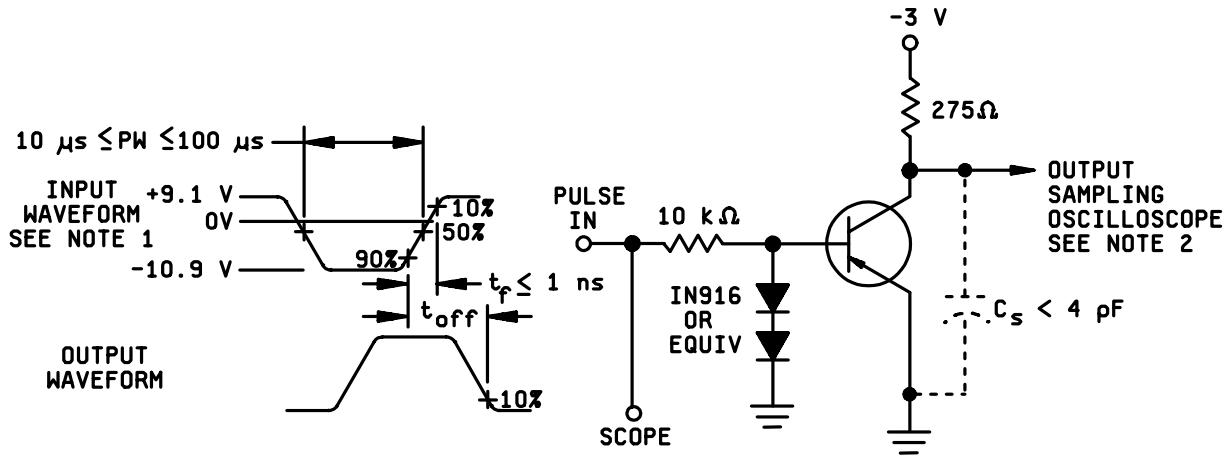
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NOTES:

1. The rise time ($t_r$) of the applied pulse shall be ≤ 1.0 ns, duty cycle ≤ 2 percent, and the generator source Z shall be 50Ω.
2. Sampling oscilloscope: $Z_{IN}$ ≥ 100 kΩ; rise time($t_r$) ≤ .1 ns.

FIGURE 5. Delay and rise time, test circuit.




NOTES:

1. The rise time ($t_r$) of the applied pulse shall be ≤ 1.0 ns, duty cycle ≤ 2 percent, and the generator source Z shall be 50Ω.
2. Sampling oscilloscope: $Z_{IN}$ ≥ 100 kΩ; rise time ($t_r$) ≤ .1 ns.

FIGURE 6. Storage and fall time, test circuit.



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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. The acquisition requirements are as specified in MIL-PRF-19500.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers' List (QML) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43216-5000.

6.4 Suppliers of JANHC and JANKC die. The qualified JANHC/JANKC suppliers with the applicable letter version (example, JANHCA2N3250A) will be identified on the QML.

| JANC ordering information | |
|---|---|
| PIN | Manufacturer |
| | 12498 |
| 2N3250A<br>2N3251A | JANHCA2N3250A<br>JANHCA2N3251A |
| 2N3250A<br>2N3251A | JANKCA2N3250A<br>JANKCA2N3251A |



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6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - NW
Air Force - 11
NASA - NA
DLA - CC

Review activities:
Army - AR, AV, MI, SM
Navy - AS, MC
Air Force - 19

Preparing activity:
DLA - CC

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**STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL**

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| **I RECOMMEND A CHANGE:** | 1. DOCUMENT NUMBER<br>MIL-PRF-19500/323G | 2. DOCUMENT DATE<br>22 June 2001 |
|---|---|---|

3. **DOCUMENT TITLE**
SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, SWITCHING TYPES 2N3250A, 2N3251A, 2N3250AUB, 2N3251AUB, JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>COMMERCIAL<br>DSN<br>FAX<br>EMAIL | 7. DATE SUBMITTED |

8. PREPARING ACTIVITY

| a. Point of Contact<br>Alan Barone | b. TELEPHONE<br>Commercial: 614-692-0510 DSN: 850-0510 FAX: 614-692-6939 EMAIL: alan_barone@dscc.dla.mil |
|---|---|
| c. ADDRESS<br>Defense Supply Center, Columbus<br>ATTN: DSCC-VAC<br>Post Office Box 3990<br>Columbus, OH 43216-5000 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Standardization Program Office (DLSC-LM)<br>8725 John J. Kingman, Suite 2533<br>Fort Belvoir, VA 22060-6221<br>Telephone (703) 767-6888 DSN 427-6888 |



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